

Rahul Sharma - Work Sample Portfolio

Prepared for hiring managers and technical interview loops.

Profile: Data Scientist / AI Engineer

Experience: 5+ years in production ML, GenAI systems, and applied analytics

Core themes: Product-focused AI systems, agentic workflows, RAG, multimodal pipelines, predictive modeling, MLOps, cloud deployment, and analytics-to-decision delivery

Portfolio breadth: 12+ projects spanning healthcare AI, operations intelligence, learning systems, and multimodal ML

Report Scope and Portfolio Context

- This report highlights three flagship hosted systems: AI Cargo, Doc2Data, and PathWise.
- These projects were selected to represent applied AI delivery across operations, healthcare, and learning products.
- Additional projects are available in the portfolio repository, including Streamlit-hosted prototypes and research-focused implementations.
- The structure below is designed for recruiter and hiring-manager scanability: problem, scope, architecture, links, and impact.

Project 01 - AI Cargo

Smith Challenge · 1st Prize (\$4K)

Agentic Operations

Cold-chain Risk

Compliance AI

Overview

AI Cargo is an agentic risk intelligence platform for pharmaceutical cold-chain shipments. It combines deterministic compliance rules and ML predictions, then orchestrates mitigation workflows through tool-using agents with human-in-the-loop controls. **Winner of first prize (\$4,000) in the Smith Agentic AI Challenge.**

Problem Solved

Temperature-sensitive logistics requires immediate, compliant, and auditable decisions. Manual intervention is often slow and inconsistent for medium/high risk events.

End-to-End System Scope

1. Ingest telemetry windows from Supabase with fallback data support.
2. Engineer risk features and score windows via deterministic + XGBoost fusion.
3. Route medium/high/critical events to LangGraph orchestration.
4. Execute specialized tools (compliance, routing, scheduling, insurance, triage, approvals, notifications).
5. Surface outcomes in FastAPI + React dashboard with traceability and audit logs.

Architecture Diagram

System architecture: [View link](#)

Data flow architecture: [View link](#)

System Architecture

AI Cargo system architecture diagram preview

Data Flow

AI Cargo data flow diagram preview

Demo

Live demo: [Open application](#)

GitHub

Source code: [View repository](#)

Tech Stack

Python, FastAPI, LangGraph, XGBoost, SHAP, Optuna, Supabase, React, Tailwind.

Business Impact Narrative: Improves intervention speed for high-risk cold-chain incidents, adds compliance transparency through RAG-backed validation and logging, and demonstrates practical agentic orchestration with governed human approval.

Project 02 - Doc2Data

Multimodal AI

Document Intelligence

Healthcare Automation

Overview

Doc2Data is a healthcare document-to-JSON extraction platform that processes CMS-1500 and UB-04 forms using lane-based extraction, alignment, OCR, validation, and targeted rescue logic.

Problem Solved

Healthcare form extraction is noisy and expensive when every document is forced through the same OCR pipeline. Doc2Data adapts extraction strategy per document type and quality signal.

End-to-End System Scope

1. Upload PDF/image through Next.js frontend.
2. LangGraph identifies form type and chooses extraction lane (widgets, digital text, scan OCR, generic).
3. Alignment + OCR pipeline executes with quality gates.
4. Validation pass identifies uncertain fields.
5. Targeted VLM rescue runs only on selected fields.
6. Final output returns structured fields, metadata, validation notes, and debug traces.

Architecture Diagram

System architecture: [View link](#)

System Architecture

 Doc2Data system architecture diagram preview

Demo

Live demo: [Open application](#)

GitHub

Source code: [View repository](#)

Tech Stack

FastAPI, Next.js, LangGraph, Florence-2 OCR, VLM rescue, Docker.

Business Impact Narrative: Reduces extraction time from multi-minute runs to production-viable latency, improves reliability with lane routing, validation, and confidence-aware rescue, and produces integration-ready JSON for downstream healthcare workflows.

Project 03 - PathWise

Microsoft Foundry IQ

Career Simulator

Multi-Agent RAG

GitHub Copilot

Overview

PathWise is a full-stack AI learning and career-readiness platform built for the **Agents League · Creative Apps (GitHub Copilot)** track. Drop in a PDF or job posting and PathWise turns it into personalized learning content, a citation-backed **Career Simulator** interview, and a shareable readiness report. Grounding runs through **Microsoft Foundry IQ** (Azure AI Search) with automatic failover to Supabase pgvector.

Problem Solved

Learners and career switchers have fragmented material and no reliable path from “study content” to “role-ready in an interview.” PathWise closes that gap with integrated learning intelligence, O*NET-backed career matching, and a visible multi-agent loop (plan → retrieve → generate → verify) that judges and hiring managers can inspect in real time.

End-to-End System Scope

1. Learner uploads a PDF or pastes a job description on the Learn or Career Simulator surfaces.
2. **Foundry IQ** retrieves citation-backed evidence from an Azure AI Search knowledge base; Supabase pgvector serves as transparent failover.
3. Distiller generates summaries, quizzes, flashcards, micro-lessons, and diagnostics grounded in retrieved context.
4. **Career Simulator** orchestrates Planner, Retrieval, Interviewer, Scorer, Remediation, and Report agents over resume + target role.
5. Agent Thinking panel streams SSE timeline events with citation chips; Readiness Report exports 30/60/90 plans and top skill gaps.
6. Dashboard tracks progress, mastery, and offline eval metrics (groundedness, refusal correctness, p95 latency).

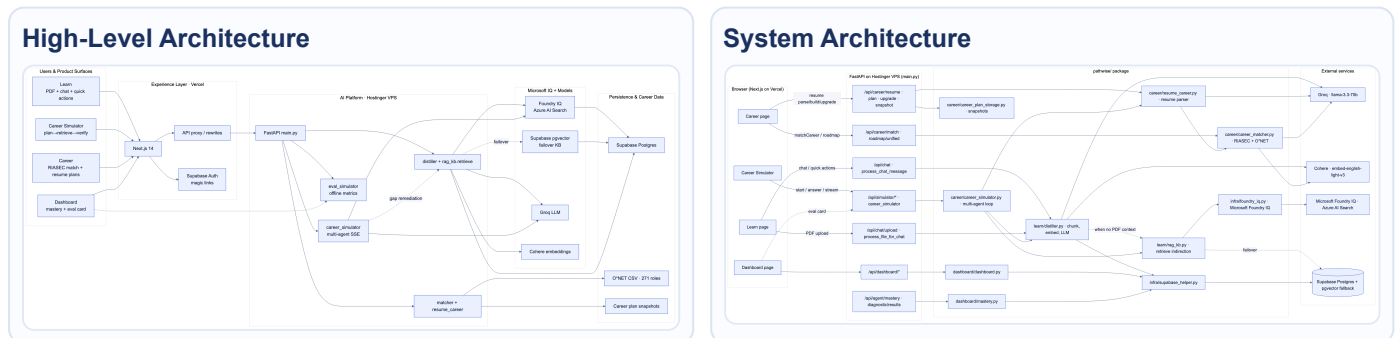
Microsoft IQ Integration

- **Foundry IQ** (Azure AI Search index `prepkb-index`) is the primary retrieval layer for Learn chat and the Career Simulator.
- Hybrid keyword + vector search with semantic reranking; every chunk carries source metadata for citations in the UI.
- Built with **GitHub Copilot** in VS Code — orchestrator, Foundry client, FastAPI routes, SSE timeline, and eval harness.

Architecture Diagram

High-level architecture: [View link](#)

System architecture: [View link](#)



Demo

Live app: [Open PathWise](#)

Career Simulator: [Open simulator](#)

GitHub

Source code: [View repository](#)

Tech Stack

Next.js 14, FastAPI, Groq (llama-3.3-70b), Cohere embeddings, **Microsoft Foundry IQ** (Azure AI Search), Supabase (auth + pgvector fallback), Docker on Hostinger VPS, Vercel frontend.

Business Impact Narrative: Converts passive content into guided, citation-backed learning; runs a demo-ready Career Simulator that chains interview, gap analysis, and roadmap in one flow; demonstrates production AI patterns (Foundry IQ grounding, provider failover, grounding guard, eval harness) suitable for enterprise and competition review.

Comparative Strength Across Work Samples

- **PathWise:** Foundry IQ-grounded learning + multi-agent Career Simulator (GitHub Copilot, Agents League).
- **AI Cargo:** Agentic operations and compliance under real-world risk constraints; first prize (\$4K), Smith Agentic AI Challenge.
- **Doc2Data:** Multimodal document AI with robust extraction and quality controls.

Together, these projects show system design depth across user-facing UX, AI orchestration, data pipelines, and production delivery.

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